

JASWANTH NARRAVULA

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SUMMARY

Software Engineer with 3+ years of experience building backend platforms, Java microservices, REST APIs, authentication systems, distributed workflows, and full-stack applications using Java 17, Spring Boot, Go, Python, SQL, and React/TypeScript. Strong in OAuth 2.1/OpenID Connect security, concurrent transaction processing, PostgreSQL/MySQL optimization, Redis caching, automated testing, CI/CD, and production reliability.

SKILLS

Languages: Java 17, Go, Python, C, C++, TypeScript/JavaScript, SQL, Bash

Backend & Microservices: Spring Boot, Spring Data JPA, Hibernate, REST APIs, FastAPI, Flask, Node.js, Microservices, OpenAPI/Swagger

Data & Storage: MySQL, PostgreSQL, Redis, Elasticsearch, SQLAlchemy, Transaction Management, Query Optimization, Indexing, Caching

Distributed Systems: Concurrent Programming, Java Multithreading, Event-Driven Architecture, Idempotency, Retry/Backoff, Horizontal Scaling, Fault Tolerance

Security & Identity: OAuth 2.1, OpenID Connect, PKCE, JWT, RS256/JWKS, Argon2id, RBAC, SSO, Microsoft Entra ID, Rate Limiting, CSRF

Engineering Tools: Maven, JUnit 5, Mockito, Postman, Git, GitHub Actions, CI/CD, Docker, Selenium, Linux

Frontend: React.js, TypeScript, Redux, Vite, Tailwind CSS

PROFESSIONAL EXPERIENCE

Xclusive Trading Inc., Sugar Land, TX: Software Engineer

Jun 2025 - Present

- Engineered an OAuth 2.1 / OpenID Connect identity provider in Go with Authorization Code + PKCE, refresh-token rotation, OIDC discovery, RS256 JWT/JWKS rotation, Argon2id, and RBAC for centralized SSO supporting 3,000+ users.
- Hardened authentication with rate limiting, CSRF/HSTS/CSP, audit logging, impersonation-safe sessions, and geo restrictions; integrated Microsoft Entra ID/OIDC with RS256 validation and route-level RBAC.
- Decomposed a production platform into 6 microservices and built idempotent, replay-safe ETL workflows with exponential backoff for reliable reprocessing without duplicate records.
- Tuned SQLAlchemy pooling/retries and PostgreSQL plans, indexes, and reporting tables, sustaining sub-200 ms p95 latency and reducing report-generation time by ~60% across 8+ services.
- Optimized React/TypeScript interfaces with code splitting, lazy loading, request deduplication, TTL caching, cancellation, and memoization; delivered WebSocket chat/notifications and standardized GitHub Actions CI/CD.

IBM, Hyderabad, India: Software Engineer

May 2022 - Dec 2023

- Engineered 10+ production REST APIs using Java 17, Spring Boot, Spring Data JPA, and Hibernate to automate corporate-loan processing, including application intake, eligibility validation, business rules, and multi-stage approvals for DBS Bank.
- Architected a state-driven workflow engine for rule-based eligibility decisions and deterministic status transitions, maintaining transactional consistency and data integrity across concurrent approval workflows backed by MySQL.
- Reduced API response times by 30% under high-concurrency workloads by analyzing SQL execution plans, optimizing MySQL queries/indexes and JPA/Hibernate access patterns, and applying Redis caching for frequently accessed data.
- Designed thread-safe service-layer components using Java concurrency and multithreading primitives to process simultaneous loan requests while preventing race conditions, inconsistent state transitions, and concurrent-update conflicts.
- Developed unit tests with JUnit 5 and Mockito and validated REST contracts with Swagger/OpenAPI and Postman; managed repeatable builds, test execution, packaging, and dependencies with Maven.

All India Council for Technical Education (AICTE) - Technical Internships, Remote

Oct 2021 - Mar 2022

- Built AWS lab environments with EC2, S3, VPC, RDS, Lambda, CloudWatch, and IAM, applying AWS Well-Architected principles across security, scalability, availability, monitoring, and cost.
- Completed applied ML labs covering data preparation, feature engineering, model training/evaluation, forecasting, computer vision, and NLP; trained and deployed models with Amazon SageMaker.

ACADEMIC PROJECTS

Price Intelligence Platform - FastAPI, PostgreSQL, Redis, Elasticsearch, Docker

- Architected a FastAPI price-intelligence platform with asynchronous Scrappy/Celery/Redis ingestion, PostgreSQL persistence, Elasticsearch fuzzy search, cross-retailer ranking, price-history analytics, and Docker Compose packaging.
- Hardened asynchronous ingestion with Celery retry and failure-handling workflows and implemented Redis TTL caching to reduce repeated PostgreSQL reads and improve reliability and search-response performance.

ASL Gesture Recognition & Translation - Python, PyTorch, ResNet-18

- Built and evaluated a PyTorch ASL recognition pipeline for 24 static gestures; transfer learning with ResNet-18 improved test accuracy from 13.21% to 43.36% and macro-F1 from 0.081 to 0.351.

EDUCATION

Master of Science in Computer Science, University of Alabama at Birmingham (UAB), Birmingham, AL

Jan 2024 - Dec 2025